Supplementary Material for "Permutations Accelerate Approximate Bayesian Computation"

S.1 Vanilla ABC Algorithm

For completeness, we recall the standard rejection ABC algorithm (**?**).

Algorithm 1. Vanilla Approximate Bayesian Computation

Input: observed dataset $\mathbf{y}$, number of iterations N, threshold $\varepsilon > 0$, summary statistic η.

Output: a sample $(\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(N)})$ from $\pi_\varepsilon\{\cdot \mid \eta(\mathbf{y})\}$.

for $i = 1, \ldots, N$ **do**

  repeat

    $\boldsymbol{\theta}^{(i)} \sim \pi(\cdot)$;

    $\mathbf{z}^{(i)} \sim f(\cdot \mid \boldsymbol{\theta}^{(i)})$;

  until $d\{\eta(\mathbf{z}^{(i)}), \eta(\mathbf{y})\}) \leq \varepsilon$;

S.2 Stratified version of permABC

Even though permABC is equivalent to classical ABC in the small ε regime, it differs outside of this regime because it integrates $\mathbf{z}$ into the constrained space $\mathcal{D}^*_{\mathbf{y}}$. Here, we propose a method called stratified permABC, which maintains the same acceptance rate but is exactly equivalent to ABC regardless of the value of ε.

In the stratified permABC algorithm, similar to the standard approach, we sample from the prior $\boldsymbol{\theta} \sim \pi(\cdot)$ and then sample data from the likelihood: $\mathbf{z} \sim f(\cdot \mid \boldsymbol{\theta})$. Here, we consider the set of all permutations in the symmetric group $\mathcal{S}_K$, and determine the subset of acceptable permutations $\mathcal{S}_K^\varepsilon = \sigma \in \mathcal{S}_K : d(\mathbf{y}, \mathbf{z}_\sigma) \leq \varepsilon$. The acceptance condition is then $|\mathcal{S}_K^\varepsilon| > 0$, which is equivalent to $\min_{\sigma \in \mathcal{S}_K} d(\mathbf{y}, \mathbf{z}_\sigma) \leq \varepsilon$. We then accept the simulated $\boldsymbol{\theta}_{\sigma'}$, where $\sigma' \sim \mathcal{U}(\mathcal{S}_K^\varepsilon)$ with a weight proportional to $|\mathcal{S}_K^\varepsilon|$.

This methodology produces samples from the classical ABC pseudo-posterior distribution:

$$
\begin{aligned}
\bar{\pi}_\varepsilon(\boldsymbol{\theta} \mid \mathbf{y}) &\propto \pi(\boldsymbol{\theta}) \int_{A_{\mathbf{y},\varepsilon}} \frac{f(\mathbf{z} \mid \boldsymbol{\theta})}{|\mathcal{S}_K^\varepsilon|} d\mathbf{z} \cdot |\mathcal{S}_K^\varepsilon| \\
&\propto \pi_\varepsilon(\boldsymbol{\theta} \mid \mathbf{y})
\end{aligned} \tag{S.1}
$$

However, this ABC scheme is impractical because it is too costly. Sampling $K!$ permutations is unfeasible. Thus, we can write this last quantity $|\mathcal{S}_K^\varepsilon|$ as $\mathbb{E}_{\sigma \sim \mathcal{U}(\mathcal{S}_K)} \left[\mathbb{I}_{A_{\mathbf{y},\varepsilon}}(\mathbf{z}_\sigma) \right]$, which we will estimate by Monte Carlo. For small ε, a standard Monte Carlo simulation will have large variance: indeed, for most permutations σ, $d(\eta(\mathbf{y}), \eta(\mathbf{z}_\sigma)) > \varepsilon$, i.e., $\mathbb{I}_{A_{\mathbf{y},\varepsilon}}(\mathbf{z}_\sigma) = 0$.

Instead, we resort to importance sampling. We write

$$
\mathbb{E}_{\sigma \sim \mathcal{U}(\mathcal{S}_K)} \left[\mathbb{I}_{A_{\mathbf{y},\varepsilon}}(\mathbf{z}_\sigma) \right] = \mathbb{E}_{\sigma \sim \rho} \left[\frac{\mathbb{I}_{A_{\mathbf{y},\varepsilon}}(\mathbf{z}_\sigma)}{K! \rho(\sigma)} \right]
$$

To do this, we sample $\sigma_l \sim \rho(\sigma)$ for $l = 1, \ldots, L$ and replace the weights $|\mathcal{S}_K^\varepsilon|$ by the following importance sampling estimator: $\frac{1}{L} \sum_{l=1}^L \frac{\mathbb{I}_{A_{\mathbf{y},\varepsilon}}(\mathbf{z}_{\sigma_l})}{\rho(\sigma_l)}$.

We now look for a good proposal distribution ρ over $\mathcal{S}_K$.

S.2.1 Simulating Non-uniform Permutations

A good proposal π will have support all of the symmetric group $\mathcal{S}_K$, but will be concentrated on values which are more likely to verify σ, $d(\eta(\mathbf{y}), \eta(\mathbf{z}_\sigma)) < \varepsilon$. We proceed in two steps:

- We compute $\sigma^* = \arg\min_\sigma d(\eta(\mathbf{y}), \eta(\mathbf{z}_\sigma))$, which can be done efficiently using a Jonker-Volgenant algorithm. (Complexity $\mathcal{O}(K^3)$)

- If $d(\eta(\mathbf{y}), \eta(\mathbf{z}_{\sigma^*})) > \varepsilon$, we must reject $\boldsymbol{\theta}$. On the other hand, if $d(\eta(\mathbf{y}), \eta(\mathbf{z}_{\sigma^*})) < \varepsilon$, we use a distribution $\rho(\cdot \mid \sigma^*)$ concentrated near σ^*.

We now propose a strategy to devise such a distribution $\rho(\cdot \mid \sigma^*)$. An additional constraint is that since this distribution will be used in importance sampling, its probability mass function must be computable pointwise. Our distribution is best understood through its generative process:

1. Generate an auxiliary variable N from a distribution p_N with support $\{1, \dots, K\}$; we choose the arbitrary distribution $N \sim 1 + \text{Poisson}(\lambda)$ truncated at K.

2. If $N = 1$ then return $\sigma = \sigma^*$.

3. If $N > 1$, generate σ uniformly from the set $\mathfrak{S}_N = \{\sigma \in \mathcal{S}_K : \|\sigma - \sigma^*\|_0 = N\}$ where $\|\cdot\|_0$ is the Hamming or edit distance: $\|\sigma - \sigma^*\|_0 = \sum_{j=1}^{K} \mathbb{I}_{\{\sigma(i) \neq \sigma^*(i)\}}$.

Note that this choice is valid because $\cup_n \mathfrak{S}_n = \mathcal{S}_K$ and because $\mathfrak{S}_1 = \emptyset$ and $\mathfrak{S}_0 = \{\sigma^*\}$.

To generate uniformly from $\mathfrak{S}_n$, we select uniformly n elements out of K, generate a random derangement of length n, and apply it to the selected elements. Recall that a derangement is a permutation that leaves no point invariant; the number of derangements of length n is called the subfactorial of n, denoted by $!n$ and is easily computed: $!n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!} = \lceil \frac{n!}{e} \rfloor$ where $\lceil \cdot \rfloor$ denotes the closest integer. The set $\mathfrak{S}_n$ is known as the set of the partial derangements and is then such that $|\mathfrak{S}_n| = \binom{K}{n} !n$. Thus the probability of sample σ^* is: $\rho(\sigma^* \mid \sigma^*) = p_N(1)$ and for permutation $\sigma \neq \sigma^*$, let $n = \|\sigma - \sigma^*\|_0$, the mass function at σ is:

$$\rho(\sigma \mid \sigma^*) = p_N(n) \frac{1}{|\mathfrak{S}_n|} = p_N(n) \frac{1}{\binom{K}{n} !n}$$

.

S.2.2 Stratified estimator

With this instrumental distribution centered at σ^*, we can enhance the effectiveness of our method through the implementation of a stratified estimator. In doing so, we can manage the permutations we randomly sample, ensuring acceptance of parameter $\boldsymbol{\theta}$ if at least one permutation (specifically σ^*) satisfies the ABC conditions.

We establish a partition of the set $\mathcal{S}_K$ into H strata denoted as (D_h) with corresponding weights (w_h) for $h = 1, \ldots, H$. Here, $D_1 = \mathfrak{S}_0 = \{\sigma^*\}$, $D_h = \mathfrak{S}_h$ for $h = 2, \ldots, H-1$, and finally $D_H = \cup_{h=H}^{K} \mathfrak{S}_h$, with $w_h = p_N(h)$ for $h = 1, \ldots, H-1$ and $w_H = \sum_{h'=H}^{K} p_N(h')$. Subsequently, we define a new sampling distribution as a mixture across our strata: $\rho(\sigma) = \sum_{h=1}^{H} w_h \cdot \rho_h(\sigma)$, where ρ_h represents the distribution within each stratum D_h: $\rho_h = \mathcal{U}(D_h)$ for $h = 1, \ldots, H-1$, and $\rho_H = \frac{1}{w_H} \sum_{h=H}^{K} p_N(D_h) \cdot \mathcal{U}(\mathfrak{S}_h)$. Additionally, we select the positive number $L_h > 0$ of permutations within each stratum. Thus, for the initial stratum, we must have $L_1 = 1$, ensuring consideration of σ^*, and $L_h > 0$ for $h = 2, \ldots, H$. Consequently, we derive the following stratified estimator with $\sigma_l^{(h)} \sim \rho_h$ for $h = 1, \ldots, H$ and $l = 1, \ldots, L_h$:

$$\mathbb{E}_{\sigma \sim \mathcal{U}(\mathcal{S}_K)} \left[\mathbb{I}_{A_{\mathbf{y},\varepsilon}}(\mathbf{z}_\sigma) \right] \approx \sum_{h=1}^{H} \frac{1}{L_h} \sum_{l=1}^{L_h} \frac{\mathbb{I}_{A_{\mathbf{y},\varepsilon}}(\mathbf{z}_{\sigma_l^{(h)}})}{\rho_h(\sigma_l^{(h)})}$$

In this section, we have presented an effective ABC simulation method that seamlessly integrates with the traditional accept-reject ABC Algorithm, often referred to as ABC Vanilla. This integration leads to the formulation of Algorithm **??**, which we specifically label as permABC Vanilla.

S.3 Theoretical Guarantees for permABC

We now provide the proofs of the theoretical results stated in Section **??**. The key insight is that for small enough ε, the optimal permutation is unique, which implies that permABC and standard ABC accept the same samples.

Algorithm 2. Stratified permABC Vanilla

Input: observed dataset $\mathbf{y}$, number of iterations N, threshold $\varepsilon > 0$, number of strata H, proposal distributions $\rho_1, \ldots, \rho_H$.

Output: a sample $\{\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(N)}\}$ from the permuted ABC approximation $\pi_\varepsilon(\cdot \mid \mathbf{y})$.

for $i = 1, \ldots, N$ **do**

  repeat

    $\boldsymbol{\theta} \sim \pi(\cdot)$;

    $\mathbf{z} \sim f(\cdot \mid \boldsymbol{\theta})$;

    $\sigma^* = \arg\min_\sigma d(\mathbf{y}, \mathbf{z}_\sigma)$;

    $d_i^* = d(\mathbf{y}, \mathbf{z}_{\sigma^*})$;

  until $d(\mathbf{y}, \mathbf{z}_{\sigma^*}) \leq \varepsilon$;

  for $h = 1, \ldots, H$ **do**

    Generate permutations $\sigma_1^{(h)}, \ldots, \sigma_{L_h}^{(h)} \sim \rho_h(\cdot)$;

    for $l = 1, \ldots, L_h$ **do**

      Compute $d_l^{(h)} = d(\mathbf{y}, \mathbf{z}_{\sigma_l^{(h)}})$;

      Compute weight $\omega_l^{(h)} = \frac{\mathbb{1}\{d_l^{(h)} < \varepsilon\}}{\rho_h(\sigma_l^{(h)})}$;

  Normalize all weights: $\tilde{\omega}_l^{(h)} = \omega_l^{(h)} / \sum_{h=1}^{H} \sum_{l=1}^{L_h} \omega_l^{(h)}$;

  Sample σ^{prop} among $\{\sigma_l^{(h)}\}_{l,h}$ with probability $\tilde{\omega}_l^{(h)}$;

  Set $\boldsymbol{\theta}^{(i)} = \boldsymbol{\theta}_{\sigma^{\text{prop}}}$;

  Assign weight $W_i \propto \sum_{h=1}^{H} \frac{1}{L_h} \sum_{l=1}^{L_h} \frac{\mathbb{1}\{d(\mathbf{y}, \mathbf{z}_{\sigma_l^{(h)}}) < \varepsilon\}}{\rho_h(\sigma_l^{(h)})}$;

of Lemma **??**. Let $\sigma^* := \arg\min_{\sigma \in \mathcal{S}_K} d(\mathbf{y}, \mathbf{z}_\sigma)$ and assume $\varepsilon < \varepsilon^*$. We consider two cases:

Case 1: If $d(\mathbf{y}, \mathbf{z}_{\sigma^*}) > \varepsilon$, then no permutation satisfies the ABC condition:

$$\operatorname{Card}\left(\{\sigma \in \mathcal{S}_K : d(\mathbf{y}, \mathbf{z}_\sigma) \leq \varepsilon\}\right) = 0.$$

Case 2: Suppose without loss of generality that $\sigma^* = \mathrm{Id}$ and $d(\mathbf{y}, \mathbf{z}) \leq \varepsilon$. For any $\sigma \neq \mathrm{Id}$, we have:

$$\begin{aligned} d(\mathbf{y}, \mathbf{z}_\sigma) &\geq |d(\mathbf{y}, \mathbf{y}_\sigma) - d(\mathbf{y}_\sigma, \mathbf{z}_\sigma)| \\ &= |d(\mathbf{y}, \mathbf{y}_\sigma) - d(\mathbf{y}, \mathbf{z})| \\ &> \varepsilon, \end{aligned}$$

since $d(\mathbf{y}, \mathbf{y}_\sigma) > 2\varepsilon$ and $d(\mathbf{y}, \mathbf{z}) \leq \varepsilon$. Therefore, only the identity permutation satisfies the ABC criterion. □

of Theorem **??**. Let $\varepsilon < \varepsilon^*$. For every simulated dataset $\mathbf{z} \in B_\varepsilon(\mathbf{y})$ such that $d(\mathbf{y}, \mathbf{z}) \leq \varepsilon$, Lemma **??** implies that at most one permutation satisfies $d(\mathbf{y}, \mathbf{z}_\sigma) \leq \varepsilon$. In that case, the optimal permutation is necessarily $\sigma = \mathrm{Id}$.

Consequently, permABC and standard ABC both accept the same set of simulations, and:

$$A^*_{\mathbf{y},\varepsilon} = B_\varepsilon(\mathbf{y}) \cap \mathcal{D}^{*K}_{\mathbf{y}} = B_\varepsilon(\mathbf{y}) = A_{\mathbf{y},\varepsilon},$$

which implies:

$$\pi^*_\varepsilon(\cdot \mid \mathbf{y}) = \pi_\varepsilon(\cdot \mid \mathbf{y}).$$

□

of Corollary **??**. By Theorem **??**, we have $\pi^*_\varepsilon(\cdot \mid \mathbf{y}) = \pi_\varepsilon(\cdot \mid \mathbf{y})$ for all $\varepsilon < \varepsilon^*$. If $\pi_\varepsilon(\cdot \mid \mathbf{y}) \to \pi(\cdot \mid \mathbf{y})$ as $\varepsilon \to 0$, then the same convergence holds for $\pi^*_\varepsilon(\cdot \mid \mathbf{y})$. □

S.4 Technical Details for Over-Sampling

S.4.1 Asymptotic analysis: $M \to \infty$

We provide here the full derivation of the $M \to \infty$ limit stated in Section **??**. Define

$$d_M(\mathbf{y}, \mathbf{z}_{1:M}) := \min_{\sigma \in \mathcal{A}_K^M} d(\mathbf{y}, \mathbf{z}_\sigma), \tag{S.2}$$

and, for fixed β, let

$$g_\beta(z) := \int g(z \mid \beta, \mu)\, \pi(d\mu), \qquad S_\beta := \operatorname{supp}(g_\beta). \tag{S.3}$$

We introduce the support-based discrepancy

$$\delta_\beta(\mathbf{y}) := \inf_{x_{1:K} \in S_\beta^K} d(\mathbf{y}, x_{1:K}). \tag{S.4}$$

Under mild regularity assumptions on d, if $Z_1, \ldots, Z_M \mid \beta \overset{\text{i.i.d.}}{\sim} g_\beta$, then

$$d_M(\mathbf{y}, Z_{1:M}) \xrightarrow[M\to\infty]{a.s.} \delta_\beta(\mathbf{y}). \tag{S.5}$$

Hence, for every ε away from the boundary case $\delta_\beta(\mathbf{y}) = \varepsilon$,

$$\mathbb{P}_\beta\Big(d_M(\mathbf{y}, Z_{1:M}) \leq \varepsilon\Big) \xrightarrow[M\to\infty]{} \mathbf{1}\{\delta_\beta(\mathbf{y}) \leq \varepsilon\}\,. \tag{S.6}$$

Therefore the marginal Over-Sampling pseudo-posterior on the global parameter satisfies

$$\tilde{\pi}_\varepsilon^M(\beta \mid \mathbf{y}) \propto \pi(\beta)\, \mathbb{P}_\beta\Big(d_M(\mathbf{y}, Z_{1:M}) \leq \varepsilon\Big) \xrightarrow[M\to\infty]{} \pi(\beta)\, \mathbf{1}\{\delta_\beta(\mathbf{y}) \leq \varepsilon\}\,. \tag{S.7}$$

In particular, if $\delta_\beta(\mathbf{y}) = 0$ for every β in the support of the prior, then for every $\varepsilon > 0$,

$$\tilde{\pi}_\varepsilon^M(\beta \mid \mathbf{y}) \xrightarrow[M\to\infty]{} \pi(\beta). \tag{S.8}$$

Thus, infinite Over-Sampling destroys all information on the global parameter: the ABC acceptance event only measures whether the observed data lie in the support of the local marginal model. By contrast, the pushforward map $\Psi_{\mathbf{y}}^{(M)}$ retains the K matched local parameters $\mu_{\sigma^*(1:K)}$, which become concentrated around good matching values even when the marginal in β is nearly flat. As M decreases back to K, this degeneracy disappears and π_ε^{*M} connects back to the standard permABC pseudo-posterior. This $M \to \infty$ regime is qualitatively different from—and does not commute with—the large-K Wasserstein-style asymptotics of Section ??: the inferential target is recovered only at $M = K$, and Over-Sampling is therefore best viewed as an algorithmic bridge rather than an asymptotic regime to push.

S.4.2 Formal Definition of π_ε^{*M}

Given a dataset $\mathbf{y} \in \mathcal{D}^K$, the Over-Sampling acceptance condition selects particles whose associated simulated data $\mathbf{z} \in \mathcal{D}^M$ satisfy the constraint:

$$\min_{\sigma \in \mathcal{A}_K^M} d(\mathbf{y}, \mathbf{z}_\sigma) \leq \varepsilon.$$

We denote the acceptance region by

$$\tilde{A}_{\varepsilon,\mathbf{y}}^M = \left\{ \mathbf{z} \in \mathcal{D}^M \mid \min_{\sigma \in \mathcal{A}_K^M} d(\mathbf{y}, \mathbf{z}_\sigma) \leq \varepsilon \right\}.$$

The associated joint pseudo-posterior is then defined as:

$$\tilde{\pi}_\varepsilon^M(\boldsymbol{\theta}, \mathbf{z} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) f(\mathbf{z} \mid \boldsymbol{\theta}) \mathbb{1}_{\tilde{A}_{\varepsilon,\mathbf{y}}^M}(\mathbf{z}).$$

To restore identifiability, we apply a projection $\varphi_\mathbf{y}$ that reorders the parameters and data. Specifically, the optimal partial permutation $\sigma^* \in \mathcal{A}_K^M$ is used to assign the best-matching K simulated compartments to the observed data. The remaining $M - K$ components are sorted by index. We define:

$$\varphi_\mathbf{y}(\boldsymbol{\theta}, \mathbf{z}) := (\boldsymbol{\theta}_{\sigma^*}, \mathbf{z}_{\sigma^*}),$$

and the projected joint pseudo-posterior becomes:

$$\pi_\varepsilon^{*M}(\boldsymbol{\theta}, \mathbf{z} \mid \mathbf{y}) := \varphi_{\mathbf{y}\sharp} \tilde{\pi}_\varepsilon^M(\boldsymbol{\theta}, \mathbf{z} \mid \mathbf{y})$$

This distribution lives on $\Theta^M \times \mathcal{D}^M$, but we are primarily interested in the marginals over the global parameter β and the local parameters $\mu_{1:K}$ corresponding to the optimally matched compartments.

S.4.3 Choosing ε Given M_0

As discussed in Section **??**, the acceptance condition (**??**) becomes easier to satisfy as M increases. Consequently, for a given tolerance ε, a larger value of M results in a higher acceptance probability. To ensure that a sufficient number of particles are alive in the initial iteration, one must jointly select M_0 and ε so as to reach a desired acceptance rate.

Empirically, the optimal assignment distance $\min_\sigma d(\mathbf{y}, \mathbf{z}_\sigma)$ tends to decrease monotonically with M. Given a simulation budget of N particles, we define the calibration of ε as the smallest value such that at least a proportion p of the simulated datasets satisfy the condition:

$$\varepsilon := \inf \left\{ \varepsilon > 0 \mid \frac{1}{N} \sum_{i=1}^{N} \mathbb{I} \left(\min_\sigma d(\mathbf{y}, \mathbf{z}_\sigma^{(i)}) \leq \varepsilon \right) \geq p \right\}.$$

In other words, ε is chosen as the empirical quantile of order p of the distances $\min_\sigma d(\mathbf{y}, \mathbf{z}_\sigma^{(i)})$, computed on the initial sample. This guarantees that a proportion approximately p of the particles satisfy the acceptance criterion and can thus be propagated into the first population. In practice, we recommend using $p = 0.95$.

S.4.4 Designing the Sequence M_t

To avoid sudden drops in the acceptance rate between iterations, the sequence M_t must decay smoothly. A practical heuristic is to choose:

$$M_{t+1} = \lfloor K + (M_t - K)\gamma \rfloor \wedge M_t - 1$$

for some decay exponent $\gamma > 0$. This schedule allows for a slow reduction at the beginning (when population diversity is critical), followed by a sharper decay toward the final iteration where convergence to $M_T = K$ is enforced. In practice, we recommend using a conservative value such as $\gamma = 0.9$.

S.4.5 Particle Duplication Strategy

Even under a conservative schedule such as $M_{t+1} = M_t - 1$, the probability of discarding particles during the transition from iteration t to $t+1$ can remain high. This is due to the potential exclusion of compartments that previously matched the observed data under the acceptance criterion (**??**).

More precisely, the probability that a matched compartment from iteration t is excluded from the optimization at iteration $t+1$ can be upper bounded by:

$$1 - \frac{(M_t - K)!}{(M_{t+1} - K)!} \cdot \frac{M_{t+1}!}{M_t!},$$

which equals $K/(K+1)$ in the case $M_t = K+1$ and $M_{t+1} = K$. This bound highlights the non-negligible risk of mass extinction, especially in the final iterations.

To mitigate this, we introduce a duplication mechanism. Before evaluating the acceptance condition at iteration $t+1$, each accepted particle from iteration t is replicated R times using independent random partial permutations $\sigma_1, \ldots, \sigma_R \in \mathcal{A}_K^{M_{t+1}}$, where we may assume $\sigma_1 = \mathrm{Id}$ for simplicity. For each duplicated version, we test the acceptance criterion on $\mathbf{z}_{\sigma_r}$ for $r = 1, \ldots, R$.

This procedure increases the likelihood that at least one configuration of the particle will pass the acceptance test, without requiring additional simulations of the likelihood. However, it does incur additional computational cost due to the R assignment problems per particle. In practice, we find that values such as $R = 5$ or $R = 10$ are sufficient to ensure stability while keeping the cost reasonable. The value of R can be chosen adaptively based on the upper bound described above.

S.5 Technical Details for Under-Matching

S.5.1 Distance, Acceptance Region, and Projection

As introduced in Section ??, the under matching strategy accepts simulated datasets $\mathbf{z} \in \mathcal{D}^K$ when a subset of $L < K$ compartments can be matched to observed data. The acceptance region is defined as:

$$\tilde{A}^L_{\varepsilon,\mathbf{y}} = \left\{ \mathbf{z} \in \mathcal{D}^K \mid \min_{\sigma, \tilde{\sigma} \in \mathcal{A}^K_L} d(\mathbf{y}_{\tilde{\sigma}}, \mathbf{z}_\sigma) \leq \varepsilon \right\}.$$

The pseudo-posterior then takes the form:

$$\tilde{\pi}^L_\varepsilon(\boldsymbol{\theta}, \mathbf{z} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) f(\mathbf{z} \mid \boldsymbol{\theta}) \, \mathbb{1}_{\tilde{A}^L_{\varepsilon,\mathbf{y}}}(\mathbf{z}).$$

As in the Over-Sampling case, we define a projection operator $\varphi_{\mathbf{y}}$ that maps each accepted pair $(\boldsymbol{\theta}, \mathbf{z})$ to a reordered version where the L optimally matched components are moved to the top L positions, and the remaining $K - L$ components follow in order. This projected distribution, denoted π^{*L}_ε, focuses on inference for the global parameter and the best-matching local components.

S.5.2 Designing the Sequence L_t

To guide the algorithm toward the full matching regime, we define a growing schedule:

$$L_{t+1} = \lfloor L_t + (K - L_t) \cdot (1 - \gamma) \rfloor \vee L_t + 1$$

for some smoothing exponent $\gamma \in (0, 1]$. This ensures that the number of matched compartments increases gradually toward K, with smaller changes in the early iterations and a final push to full matching.

In practice, values such as $\gamma = 0.8$ or 0.9 offer a good trade-off between gradual adaptation and final convergence. The stopping point of the algorithm is $L_T = K$, at which point the standard permABC acceptance condition is recovered.

S.6 Blockwise Metropolis-within-Gibbs Kernel

This section details the MCMC transition kernel used in the permABC-SMC algorithm. At each iteration, the algorithm performs both a global move and a local move, ensuring comprehensive exploration of the posterior while maintaining computational efficiency through partial updates.

The global move proposes a new value for the shared parameter β using a symmetric random walk proposal kernel q_t, as defined in Section **??**. Synthetic data are then generated for all compartments using the proposed value β^*, and the optimal permutation minimizing the distance to the observed data is computed. The proposal is accepted or rejected based on the ABC Metropolis–Hastings criterion, using the full data discrepancy.

The local move updates the collection of local parameters $\mu_1, \ldots, \mu_K$ blockwise. The set of compartments is randomly partitioned into H equally sized blocks. Within each block, new parameters are proposed using the same Gaussian random walk kernel q_t, and new synthetic data are generated only for the updated compartments. The new configuration is accepted or rejected based on the global ABC condition, ensuring coherence of the posterior sample across all compartments.

Performing both global and local moves at every iteration leads to a total of $2NK$ data simulations per iteration for a particle population of size N, and requires $N(1 + H)$ linear assignment resolutions: one for the global move, and H for the local updates.

Although this construction is reminiscent of ABC-Gibbs methods, such as Clarté et al. (2021), the key distinction lies in the evaluation of the ABC discrepancy. In our case, the acceptance criterion remains global—defined over the full dataset—regardless of whether global or local parameters are being updated. This ensures that accepted configurations correspond to coherent fits to the observed data across all compartments.

In the Over-Sampling setting, we consider a total of $M > K$ compartments. At iteration

t, the kernel operates on the current set of M_t simulated compartments, among which only K are matched to the observed data under the optimal injection. These K matched compartments are partitioned into H blocks for Gibbs updates using the kernel q_t, while the remaining $M_t - K$ unmatched compartments form a separate block, updated independently by redrawing from the prior. Components beyond M_t are frozen.

In the Under-Matching setting, the kernel operates on the current subset of $L_t < K$ matched compartments. These L_t compartments are partitioned into H blocks, each updated via random walk proposals q_t, while the $K - L_t$ unmatched compartments are handled separately and sampled directly from the prior. In both cases, only the matched components contribute to the ABC discrepancy and drive acceptance.

S.7 Comparison in Simulation Count and Wall-Clock Time

In this appendix, we complement the results presented in Sections **??** and **??** by providing a joint comparison of ABC methods in terms of both simulation count and wall-clock time. These two metrics offer complementary perspectives on computational efficiency: the former captures algorithmic convergence speed, while the latter reflects the actual computational cost, including assignment resolution and data simulation.

Figure S.1 displays the total runtime required to reach a given sliced Wasserstein-2 distance to the reference posterior, for the hierarchical Gaussian model described in Equation (**??**), measured on a single-core CPU using identical implementation settings. The conclusions are consistent with those drawn from the simulation-count analysis in Figure **??**: permutation-based strategies reach lower SW_2 levels more efficiently, and this advantage becomes increasingly significant as the number of compartments K increases.

Algorithm 3. MCMC kernel with global and local updates for permABC-SMC

Input: Current state $(\boldsymbol{\theta}, \mathbf{z}, \sigma)$, tolerance ε, number of blocks H

Output: New state $(\tilde{\boldsymbol{\theta}}, \tilde{\mathbf{z}}, \tilde{\sigma})$ targeting $\pi_\varepsilon(\cdot, \cdot \mid \mathbf{y})$

$\tilde{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}$, $\tilde{\mathbf{z}} \leftarrow \mathbf{z}$, $\tilde{\sigma} \leftarrow \sigma$

Global update:

$\beta^* \sim q_t(\cdot \mid \beta)$

Generate: $\mathbf{z}_k^* \sim g(\cdot \mid \beta^*, \mu_k)$ for $k = 1, \ldots, K$

Compute: $\sigma^* \leftarrow \arg\min_{\sigma \in \mathcal{S}_K} d(\mathbf{z}_\sigma^*, \mathbf{y})$

Draw $U_g \sim \mathcal{U}(0, 1)$

if $U_g \leq \dfrac{\pi(\beta^*) q_t(\beta \mid \beta^*, \sigma^*)}{\pi(\beta) q_t(\beta^* \mid \beta, \sigma)} \cdot \mathbb{I}_{\tilde{A}_{\mathbf{y},\varepsilon}}(\mathbf{z}^*)$ **then**

  $\tilde{\beta} \leftarrow \beta^*$

  $\tilde{\mathbf{z}} \leftarrow \mathbf{z}^*$

  $\tilde{\sigma} \leftarrow \sigma^*$

Local update:

Randomly partition $\{1, \ldots, K\}$ into H blocks $(B_1, \ldots, B_H)$

for $h = 1, \ldots, H$ **do**

  $\mathbf{z}^* \leftarrow \tilde{\mathbf{z}}$

  for $k \in B_h$ **do**

    Propose: $\mu_k^* \sim q_t(\cdot \mid \tilde{\mu}_k, \tilde{\sigma})$

    Simulate: $\mathbf{z}_k^* \sim g(\cdot \mid \tilde{\beta}, \mu_k^*)$

    Compute: $\sigma^* \leftarrow \arg\min_{\sigma \in \mathcal{S}_K} d(\mathbf{z}_\sigma^*, \mathbf{y})$

    Draw $U_h \sim \mathcal{U}(0, 1)$

    if $U_h \leq \dfrac{\pi(\mu_{B_h}^*) q_t(\tilde{\mu}_{B_h} \mid \mu_{B_h}^*, \sigma^*)}{\pi(\tilde{\mu}_{B_h}) q_t(\mu_{B_h}^* \mid \tilde{\mu}_{B_h}, \tilde{\sigma})} \cdot \mathbb{I}_{\tilde{A}_{\mathbf{y},\varepsilon}}(\mathbf{z}^*)$ **then**

      $\tilde{\mu}_{B_h} \leftarrow \mu_{B_h}^*$

      $\tilde{\mathbf{z}}_{B_h} \leftarrow \mathbf{z}_{B_h}^*$

      $\tilde{\sigma} \leftarrow \sigma^*$

Return: $(\tilde{\boldsymbol{\theta}}, \tilde{\mathbf{z}}, \tilde{\sigma})$ with $\tilde{\boldsymbol{\theta}} = (\tilde{\beta}, \tilde{\mu})$

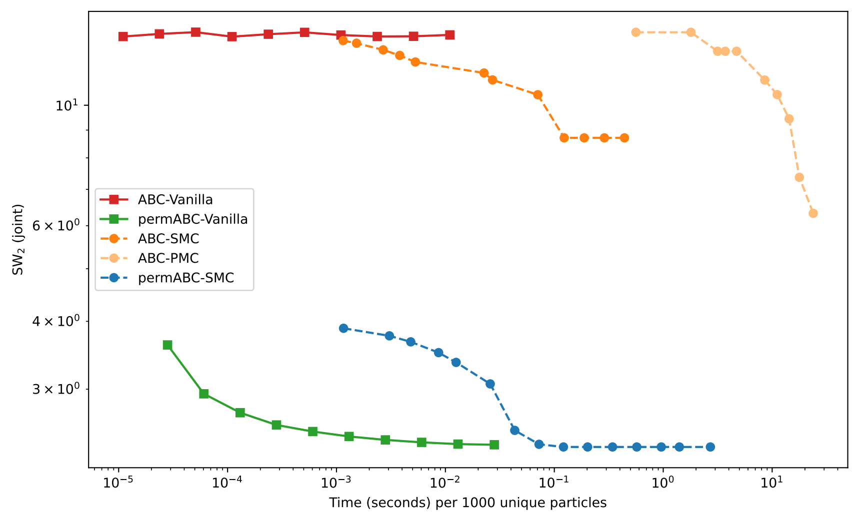

Figure S.1: Comparison of standard ABC methods and permutation-based variants on the Gaussian toy model, in terms of the sliced Wasserstein-2 distance to the reference posterior versus wall-clock time. The x-axis shows the total runtime per 1,000 unique accepted particles (log-scale), and the y-axis represents SW_2 (log-scale). Permutation-based methods consistently outperform standard ABC in terms of both accuracy and computational cost.

In the outlier-robustness setting discussed in Section ??, Figure ?? reports the wall-clock time required to reach various SW_2 levels. For completeness, we reproduce here the complementary figure comparing the same methods in terms of simulation count. The results confirm the improved sampling efficiency of Over-Sampling and Under-Matching strategies, particularly in the intermediate- and low-tolerance regimes.

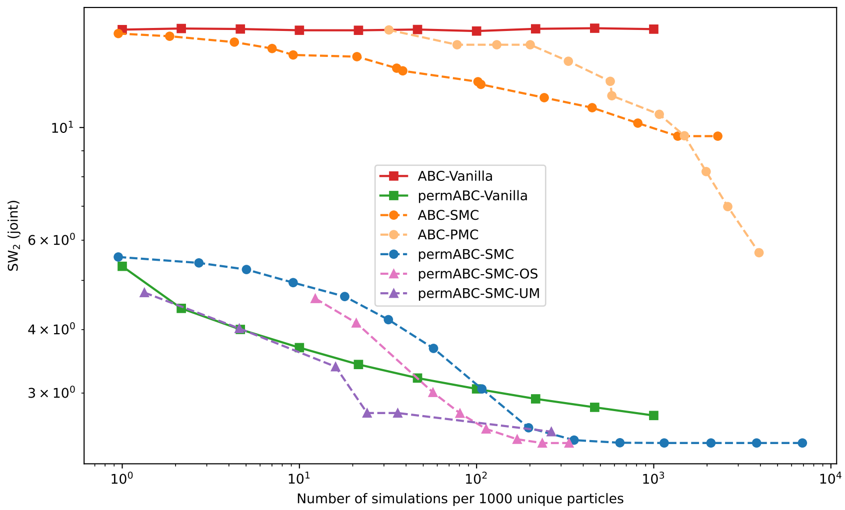

Figure S.2: Complementary comparison of ABC methods in terms of simulation count for the outlier-robustness scenario. The x-axis shows the total number of simulations per 1,000 unique accepted particles (log-scale), and the y-axis represents the sliced Wasserstein-2 distance to the reference posterior (log-scale). Over-Sampling and Under-Matching improve sampling efficiency while maintaining robustness to model misspecification.

S.8 Validation and Results of the SIR Model Application

S.8.1 Preliminary Justification for the Hierarchical Structure

Before applying permABC to the full departmental dataset, we performed a preliminary analysis to justify the modeling assumptions. In particular, we fitted independent SIR models to each department using standard ABC techniques. These runs revealed that although local dynamics exhibit variability in terms of initial conditions and intensity, the inferred values of the reproduction number R_0 were broadly consistent across departments during the first wave of the epidemic (March–July 2020).

This empirical observation supports the use of a global R_0 parameter shared across all compartments, while allowing for department-specific local parameters $\mu_k = (I_k(0), R_k(0), \delta_k)$.

The spatial distribution of δ_k reflects variations in local factors such as population density, urbanization, mobility, or healthcare capacity. These differences highlight the necessity of treating local parameters separately in a hierarchical framework, rather than aggregating data at the national level.

S.8.2 Synthetic Data Validation

To verify that permABC-SMC can recover known ground-truth parameters in the SIR setting, we generate synthetic data from the model described in Section **??**. We draw $K = 10$ compartments with true parameters sampled from realistic subregions of the prior: $R_0 \in [1.5, 3.0]$, $\nu_k \in [0.1, 0.5]$, $I_k(0) \in [1, 200]$, and $R_k(0) \in [0.1, 50]$. We then simulate $n = 120$ days of observations using the stochastic SIR simulator with log-normal noise.

We run permABC-SMC with $N = 1{,}000$ particles on the synthetic data for three values of the noise parameter: $\sigma \in \{0.01, 0.05, 0.10\}$. This serves two purposes: (i) validating that permABC-SMC recovers the true parameters when they are known, and (ii) assessing the sensitivity of the inference to the choice of σ.

Figure **??** displays the results. The top row shows the posterior of R_0 for each σ: in all three cases, the posterior is concentrated around the true value, confirming that the global parameter is well recovered. The middle row shows the marginal posteriors of four local recovery rates ν_k, each centered on the corresponding true value. The bottom row presents posterior predictive checks: 90% and 50% credible envelopes of simulated trajectories are compared to the synthetic observed data, showing good calibration.

The sensitivity to σ is as expected: smaller values produce tighter posteriors (the model is less noisy, so the data are more informative), while larger values yield wider posteriors. The choice $\sigma = 0.05$ used in the real-data application of Section **??** provides a reasonable trade-off between model flexibility and inferential precision.

References

Clarté, G., Robert, C. P., Ryder, R. J. & Stoehr, J. (2021), 'Componentwise approximate bayesian computation via gibbs-like steps', *Biometrika* **108**(3), 591–607.

fig/fig_sir_synthetic_validation_seed_42.pdf

Figure S.3: Synthetic validation of permABC-SMC on the SIR model ($K = 10$ compartments). Top: posterior of R_0 for $\sigma \in \{0.01, 0.05, 0.10\}$; dashed red line marks the true value. Middle: posteriors of four local recovery rates ν_k ($\sigma = 0.05$). Bottom: posterior predictive checks—envelopes (50% dark, 90% light) of simulated trajectories vs. synthetic observed data (solid black). The method recovers the true parameters in all cases.